



SMARTIE

Synergistic Management and Advancement of Artificial
Intelligence in European Higher Education

WP2. Alliance for Artificial Intelligence in Higher Education – Setup SMARTIE

T2.2. Analysis related to adoption AI in educational process

D.2.2.5. Analysis of the results - HEIs

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1. Foreword

The SMARTIE consortium conducted a short questionnaire with the goal of better understand the extent to which the integration of artificial intelligence (AI) into educational processes can enhance tertiary education, namely whether participants currently use AI, wish to use it more, or are sceptical about its use, their insights are invaluable.

The rapid influx of AI tools—such as ChatGPT, Turnitin, and various design applications—has prompted many in academia to reexamine traditional teaching methods and explore innovative ways to engage students and support teaching. This survey provides a multifaceted view of how institutions and educators perceive, use, and prepare for AI integration. The responses collected offer insights into:

- Geographic and institutional diversity
- Faculty background and role distribution
- Current AI tool usage and integration in teaching processes
- Perceived benefits for students and educators
- Challenges, including training, policy gaps, and trust issues
- Future strategies for institutional AI adoption

2. Introduction

The 2025 World Economic Forum Future of Jobs report showed that the technological progress is driven by artificial intelligence (AI) and Generative AI (GenAI) – and these are set to transform the labour market over the next five years.

Business leaders and professionals are feeling the impact of these rapid workplace shifts. Almost two-thirds of professional's report feeling overwhelmed by how quickly their jobs are changing. LinkedIn data shows that the jobs and skills needed are being transformed. It is observed that for every 10 professionals, at least one has a job title that didn't exist in 2000, for example data scientist or an AI engineer; 25 years ago, that wouldn't have been an option. The skills needed for work are expected to change by 70% by 2030 too, which is also a reflect of accelerated evolution and use of AI. The Future of Jobs report found that six out of every 10 business leaders expect AI and GenAI to transform their organizations, has they don't want to take the risk to be left behind.

LinkedIn has identified three core groups of skills that will be impacted by AI, and most prominently by GenAI. GenAI technologies require AI literacy skills that enable professionals to use tools such as ChatGPT and Microsoft 365 Copilot, which are already changing how people work, being AI literacy a major focus for upskilling. In the EU, for example, the number of professionals adding these skills to their profiles was 80 times greater in 2023 than in 2022. However, there are skills that GenAI does not have, like leadership, teamwork, negotiation and relationship building. These skills are critical for the implementation of AI technologies.

Effective skills policies will be those that train workers to use AI and GenAI technologies, while also supporting the development of strong people skills.

Upskilling is a priority for growth Success in AI. It is not just about investing in new technologies. There's no point in having the best AI technology if no one knows how to use it. That's why it's time to strategically design skills policies that prioritize continuous learning. Effective skills policies are key to a fast and fair AI transition. It is important to act now on AI upskilling and invest in talent programmes, so that every member of the workforce can reach their potential. This is the only way to ensure we are leveraging the full potential of AI to boost economic growth. (source: <https://www.weforum.org/stories/2025/04/linkedin-strategic-upskilling-ai-workplace-changes/>).

Based on these new job skills it is crucial to prepare HE teachers a student for what AI brings.

3. Results of the e-questionnaires

As refer before, this report examines survey data from universities and related institutions across various European countries regarding AI use in higher education. It covers institutional demographics, faculty roles and experience, current AI tool adoption, perceived benefits, readiness, challenges, and strategic recommendations. The findings reveal a cautious but optimistic attitude toward AI—faculty appreciate its potential for personalized learning and efficiency but also stress the need for clear policies, better training, and robust institutional support.

3.1 Participants Profile

3.1.1 Survey Demographics

Figure 1 indicates that most respondents are from Spain, Austria, Italy, and Romania, indicating a strong European focus in the survey. This geographic concentration suggests that the findings may be most relevant to higher education institutions (HEIs) in these countries. However, the lack of representation from other regions could limit the generalizability of the results.

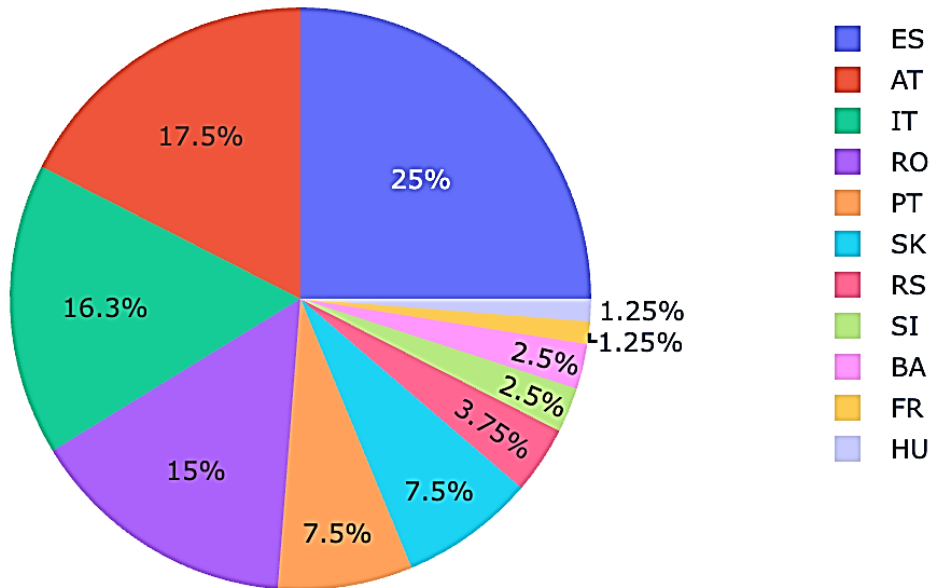


Figure 1 Country of the respondents (11 countries are represented)

3.1.2 Institution Types & Student Enrolments

Traditional universities dominate the responses, followed by applied sciences institutions. This distribution reflects a broad representation across academic structures, but the underrepresentation of vocational or specialized institutions may overlook unique challenges or opportunities these entities face in AI adoption (Figure 2).

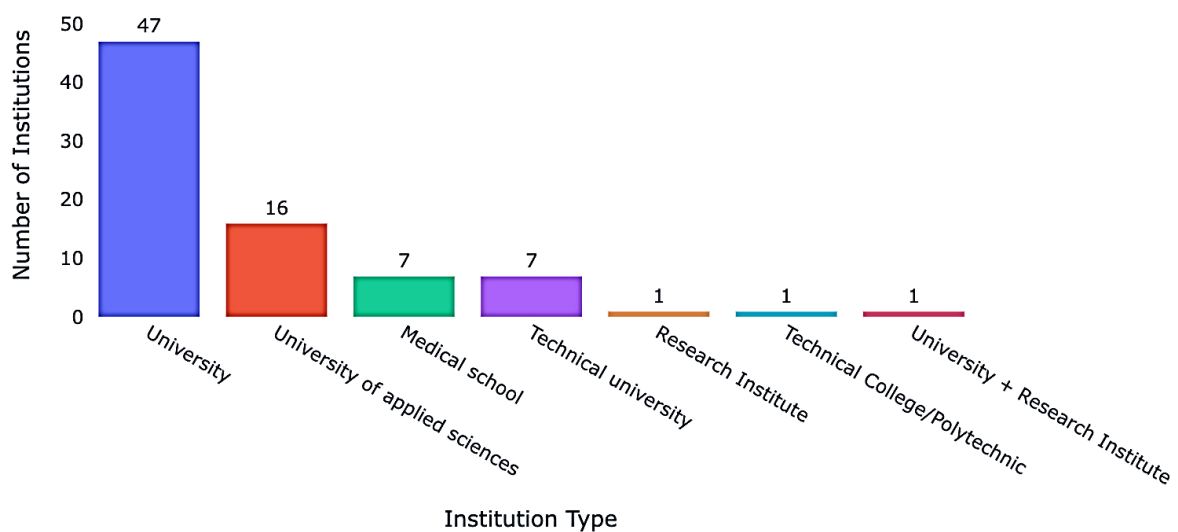


Figure 2- Institution Types

Most participating institutions are large, with enrolments exceeding 10,000 students. This suggests that AI adoption may be more feasible in well-resourced universities. However, medium and small institutions are also represented, indicating that AI integration strategies must be scalable to accommodate varying resource levels (Figure 3).

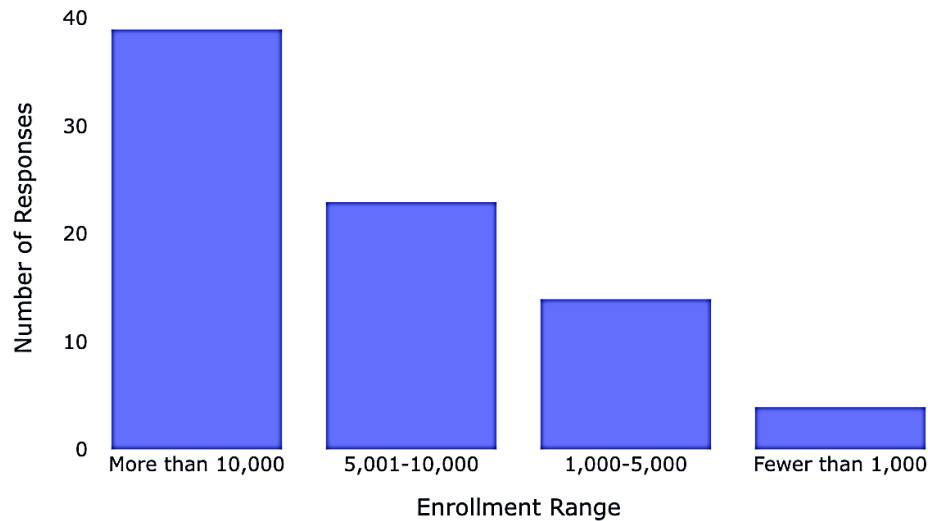


Figure 3 - Number of students per institution

3.1.3 Academic Roles, Experience, Age, and Gender

Senior academics (Associate and Full Professors) are the primary respondents, highlighting their influential role in AI adoption decisions. The limited representation of junior faculty or administrative staff may skew perspectives, as these groups might have different views on AI's practicality or ethical implications (Figure 4).

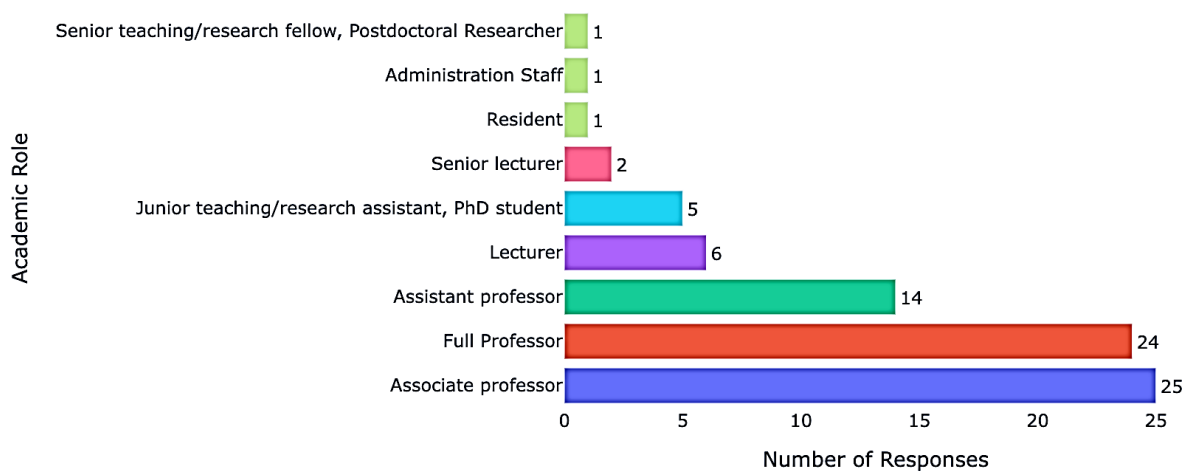


Figure 4 Distribution of Academic Roles

STEM fields (e.g., Engineering, Medicine) are most represented, aligning with higher AI adoption in technical disciplines. The lower representation from humanities or social sciences suggests a potential gap in understanding how AI can benefit non-STEM fields (Figure 5).

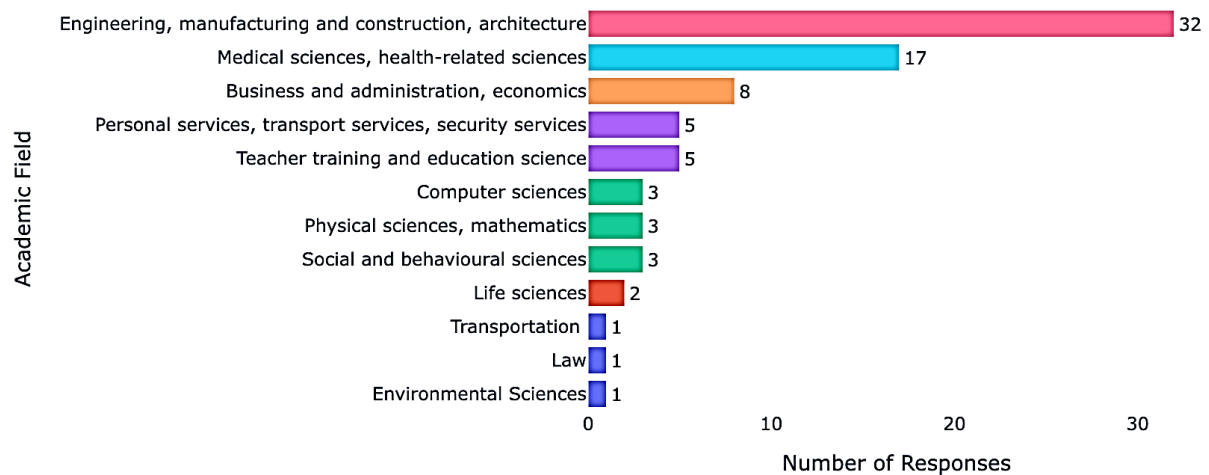


Figure 5 Distribution of Academic field

Experienced educators (11+ years) dominate, suggesting their perspectives are crucial in shaping AI policies. This result may suggest that the AI adoption must account for entrenched teaching habits. For example, tailored training for late-career educators (Figure 6).

How many years have you worked in higher education?

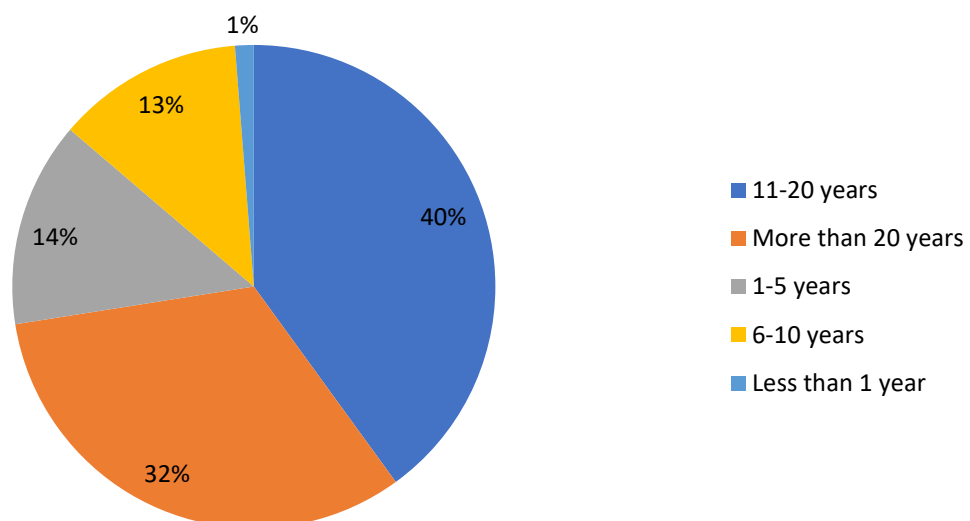


Figure 6 Academic Experience

Most respondents fall in the 46–55 age group and 36–45, with fewer younger or older faculty. This mid-career dominance may reflect a cautious optimism—experienced enough to recognize AI's potential but also aware of institutional inertia. The lack of younger voices (e.g., digital natives) might overlook innovative or disruptive perspectives (Figure 7).

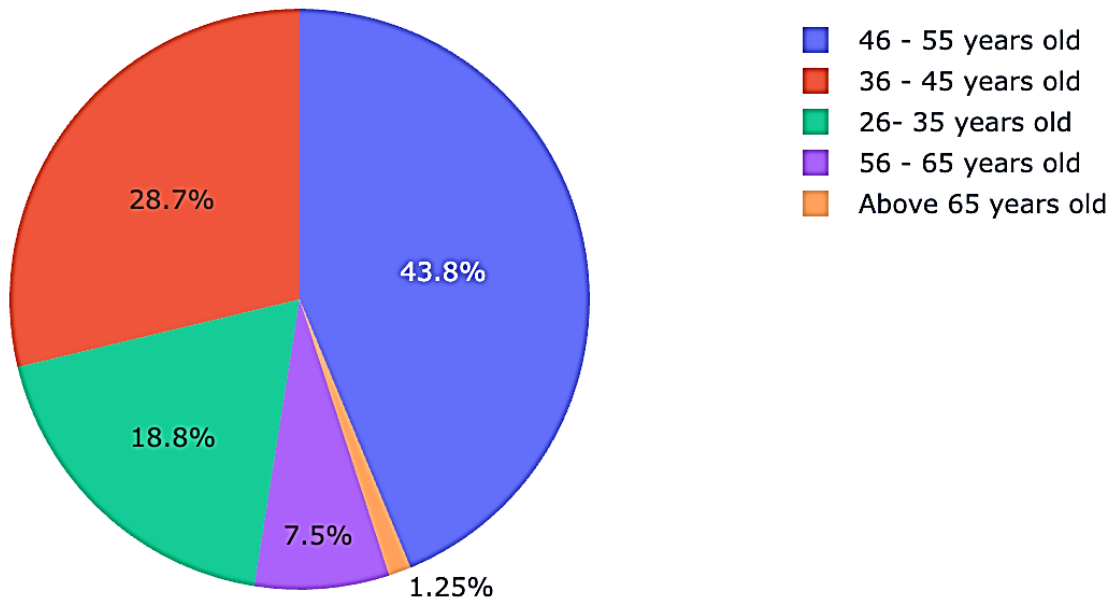


Figure 7 Age of the respondents

Male educators are the majority, indicating potential gender gaps in AI perceptions and adoption. Institutions should investigate whether this reflects broader gender disparities in STEM or leadership roles, which could influence the inclusivity of AI policies (Figure 8).

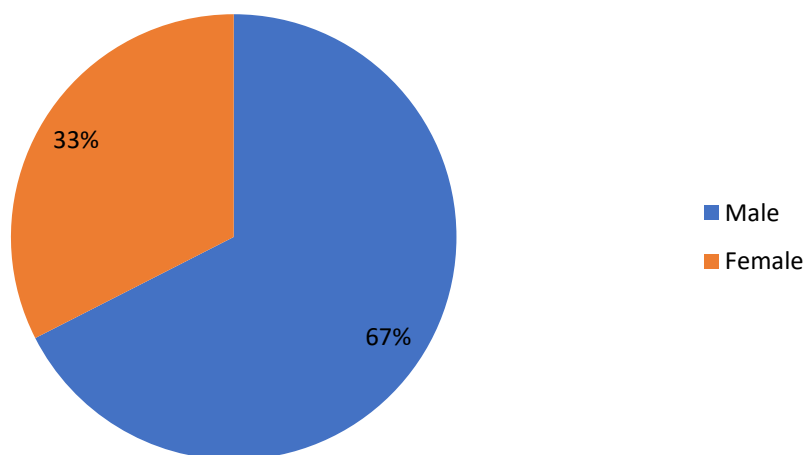


Figure 8 Gender of the respondents

While many see AI communication as positive, 35%, a notable portion remains unaffected or confused, indicating that communication causes confusion about how to use AI effectively the need for clearer messaging (Figure 9).

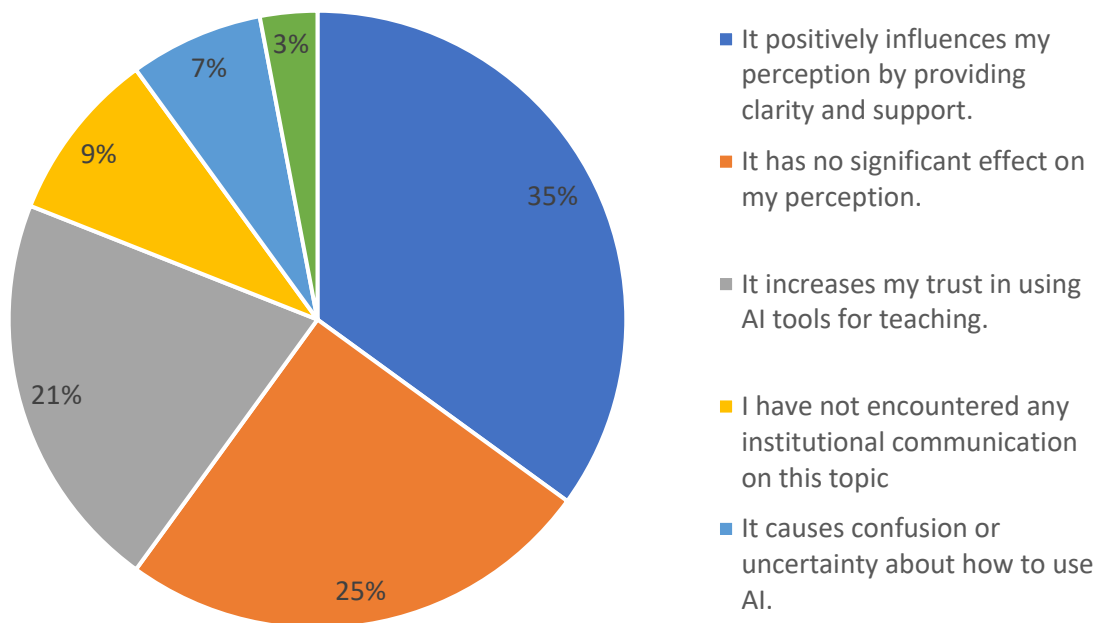


Figure 9 How institutional communication affects the perception of AI in teaching

3.1.4 In summary

Faculty respondents predominantly occupy roles such as associate and full professors with extensive experience—many having 11–20 or more than 20 years in academia. They principally come from large universities, many enrolling more than 10,000 students.

The level of experience suggests that while educators are familiar with traditional teaching methods, they face a steeper learning curve when adapting to disruptive technologies. For AI to be successfully adopted, institutions need to invest in professional development. The significant reliance on established universities also indicates that AI adoption strategies must scale to support large, complex institutions where change may be slower without clear policies and targeted training programs.

3.2 AI Adoption and Perceptions

Administrative task automation (37 responses) and adaptive learning platforms (30 responses) are seen as key benefits, freeing faculty time and personalizing learning. However, the lower emphasis on gamification (28) and 24/7 support (23) suggests that dynamic, student-centric applications of AI are still underutilized or undervalued (Figure 10).

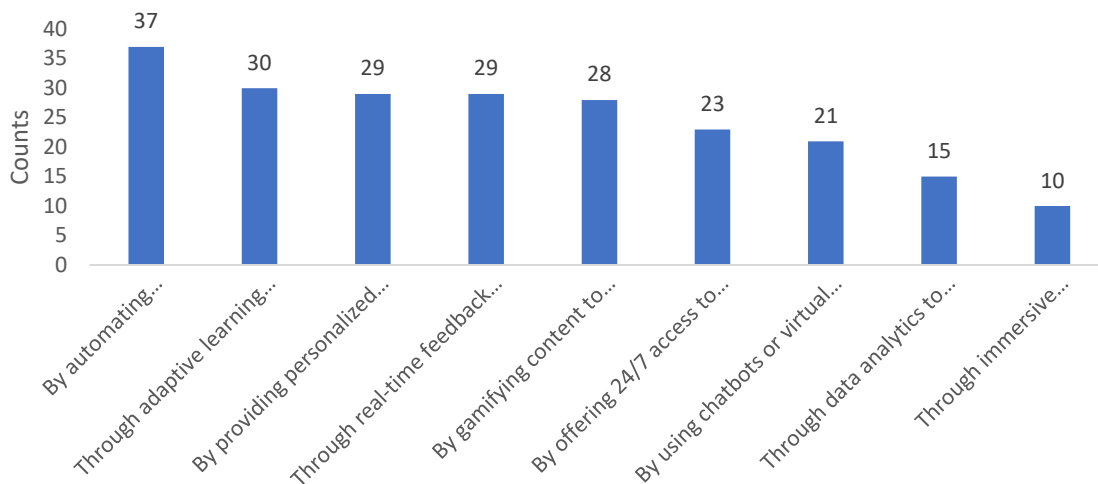


Figure 10 How can AI improve student engagement and learning outcomes

Personalizing learning materials and accessible content (47 responses each) are top priorities, alongside routine task automation (37 responses). The scepticism from a minority (3 respondents) underscores the need for demonstrable efficacy to win over hesitant educators (Figure 11).



Figure 11 How can AI support Faculty

Educators think that the major influence on 21st-century skills are the capacity of the students in analysing and evaluating the AI generated contents, and to adapt solving problems. Instant feedback to help students is also relevant, as well as the support of collaborative learning through AI platform, and once again reducing workload of teacher in

less order thinking situations. and personalizing education. So, basically, critical thinking (e.g., analysing AI-generated content) and problem-solving are highlighted as major benefits. This aligns with broader educational goals but also reveals a need for explicit training in AI literacy to ensure students can ethically and effectively leverage these tools (Figure 12).

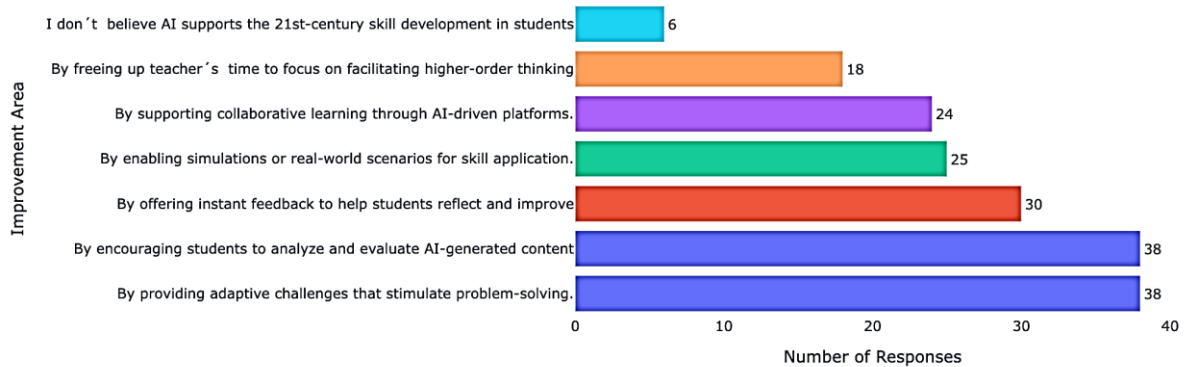


Figure 12 Ways that AI can influence the development of 21st-century skills, such as critical thinking and problem-solving

Early intervention for at-risk students and personalized learning pathways are top-cited benefits. However, the chart's lack of detail on specific analytics tools or success metrics suggests a gap between perceived potential and proven outcomes (Figure 13).

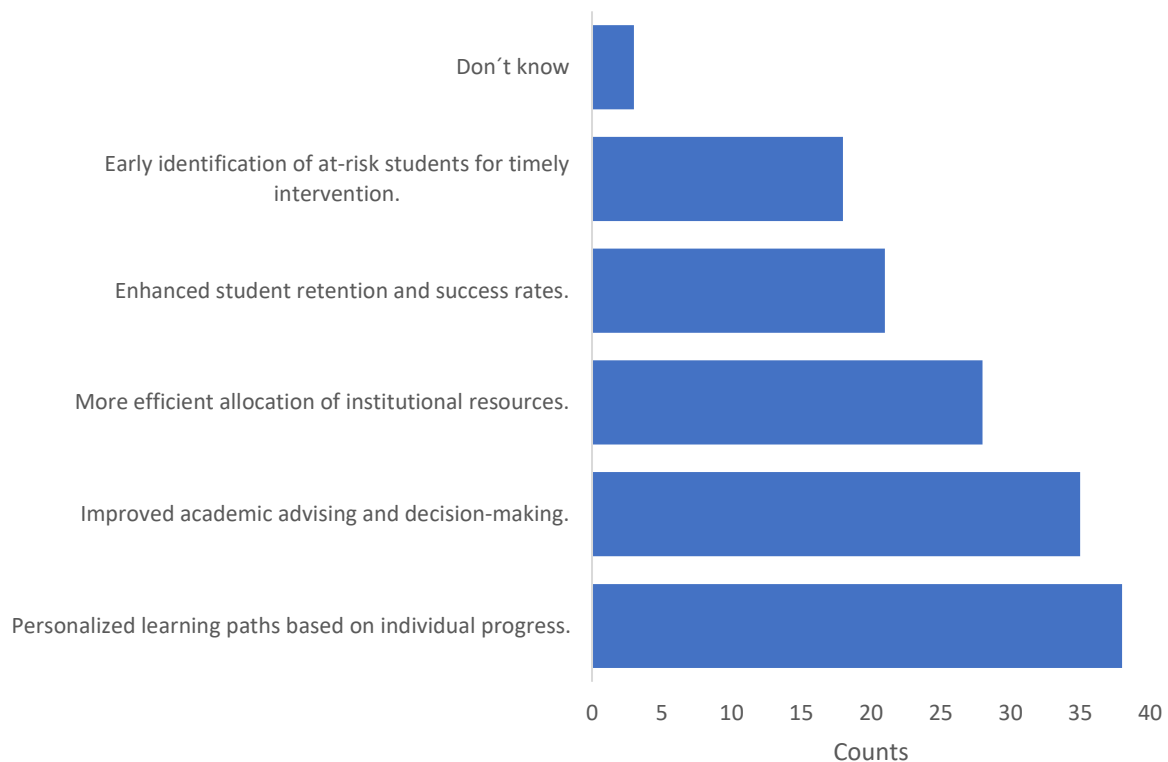


Figure 13 Potential benefits of using AI-driven predictive analytics in higher education

While many respondents agree AI can enhance engagement, the absence of qualitative data (e.g., student feedback) limits the analysis. Institutions should pair faculty perceptions with student experiences to validate these claims (Figure 14).

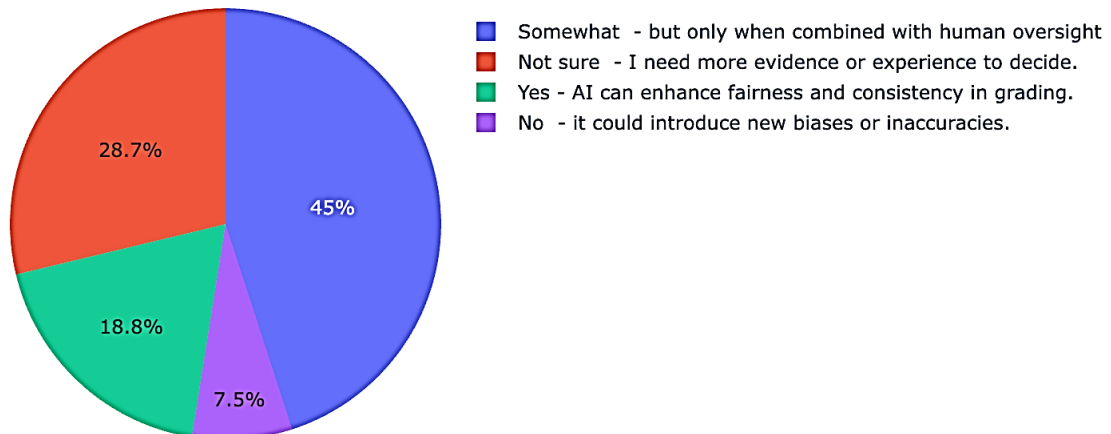


Figure 14 Can AI improve student engagement?

The chart likely reflects optimism about AI's role in quality enhancement, but without granular data, it's unclear whether improvements are anticipated in teaching, research, or administration. Further breakdowns by institutional role could clarify this (Figure 15).

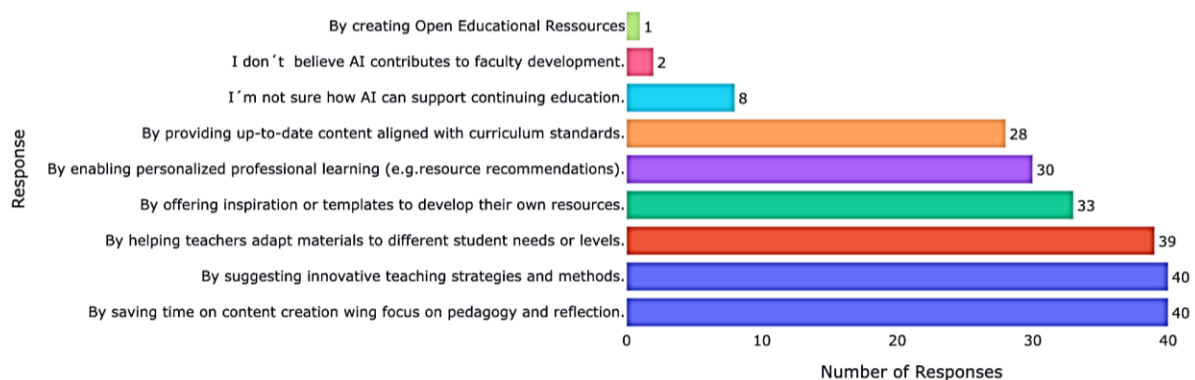


Figure 15 What is the potential of AI to improve HE quality

Responses highlight that AI is primarily used in course preparation (50 responses), with lower usage in assessment or lesson delivery. This suggests faculty are comfortable with AI as a planning aid but hesitant to delegate real-time or evaluative tasks—a trust gap that training and policy could address (Figure 16).

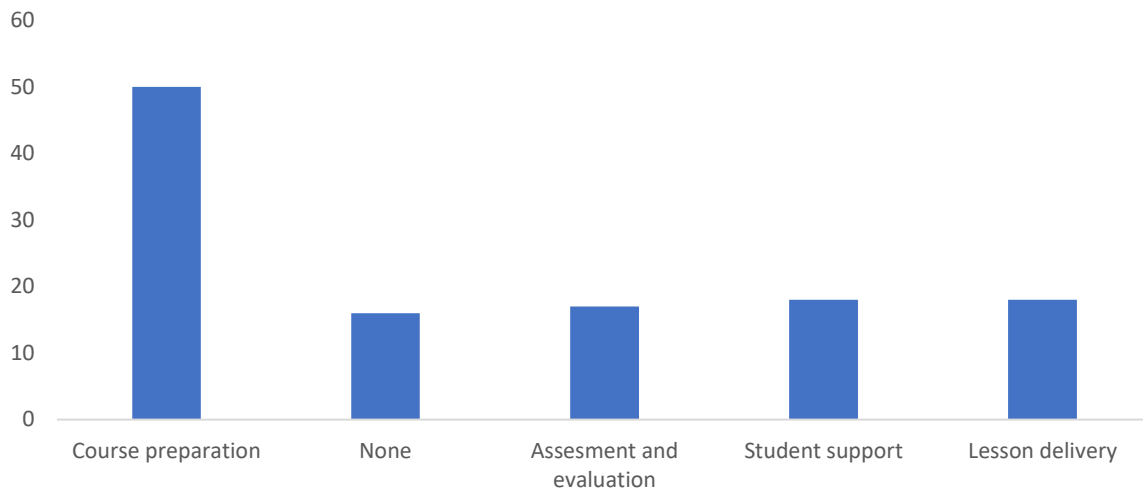


Figure 16 Stages of the teaching process where AI tools are frequently used

3.2.1 Current AI Tool Usage

There is a widespread Use of ChatGPT, which stands out with 54 mentions—indicating strong interest in conversational or content-generation AI tools. Turnitin is the next most frequently mentioned (27 responses), reflecting a focus on academic integrity and plagiarism detection. Additional tools include Canva (often mentioned with AI-driven design features) and DALL-E. The limited mention of discipline-specific tools (e.g., MATLAB for STEM) suggests a need for broader tool diversification (Figure 17).

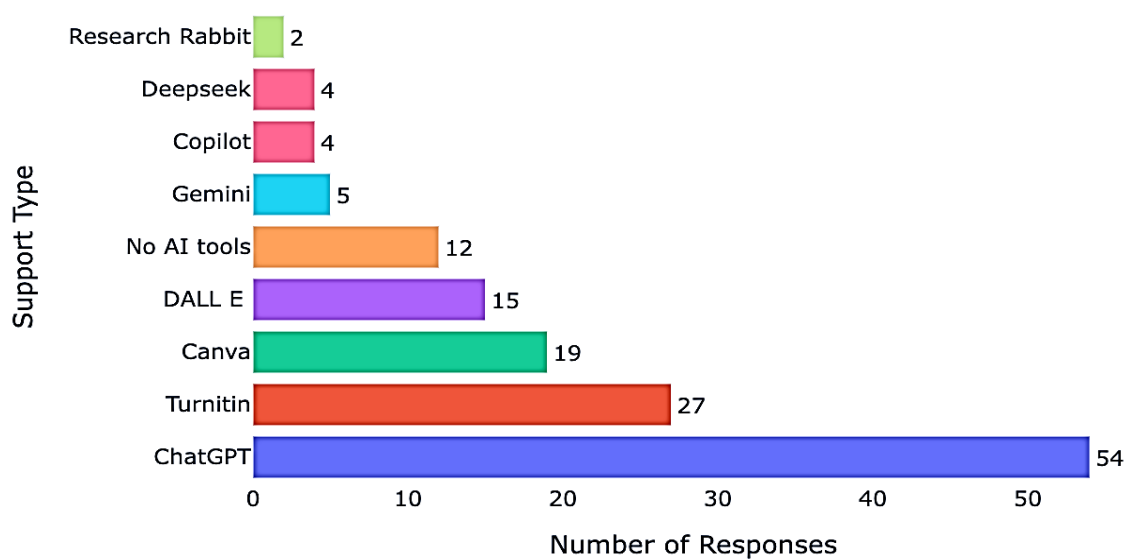


Figure 17 AI Tools usage

As for AI tools for Plagiarism detection Turnitin (23) leads, but most respondents don't use AI plagiarism tools, indicating low adoption or awareness. Institutions should evaluate whether this stems from tool limitations, cost barriers, or distrust in AI's accuracy (Figure 18).

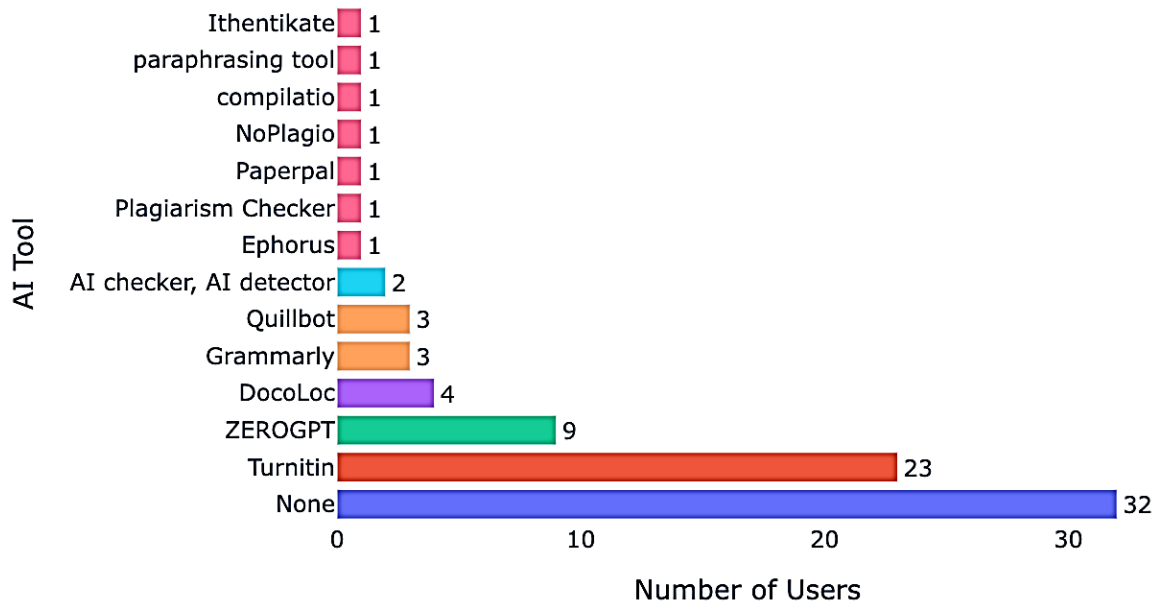


Figure 18 AI tools for Plagiarism detection

3.2.2 Summary

In terms of tools usage and Stage of Integration A high number of respondents reported using AI tools—most notably ChatGPT (54 mentions) and Turnitin (27 mentions). AI appears primarily employed during course preparation (with 50 responses), meaning that many educators see AI as a tool to generate content ideas, drafts, or to supplement curriculum planning. The predominance of AI use in course design and planning suggests that educators view AI as an assistant in the creative and administrative dimensions of teaching. However, lower usage during lesson delivery. The benefits underscore AI's potential to transform teaching through individualized learning and freeing faculty from routine tasks. When educators see AI to enhance student engagement and personalize learning pathways, they become more open to adopting it. Adoption will likely accelerate when technological tools can demonstrably improve academic advising, offer real-time data analysis, and customize learning experiences to meet student and assessment points to a cautious approach toward deploying AI for dynamic student interaction and evaluative processes. Institutions must therefore align AI integration efforts with stages where faculty already feel comfortable, gradually extending its use into real-time feedback and assessment once clear standards and training are in place.

3.3 Challenges, including training, policy gaps, and trust issues

In terms of perception of AI reliability most consider AI “somewhat reliable” and “neutral”, while a minority consider AI “very reliable” or “not at all reliable”. This mix speaks to a cautious optimism, where benefits are recognized but there is a clear need for improved outcomes and clearer operational standards (Figure 19). Clear benchmarks and transparency in AI algorithms could build trust.

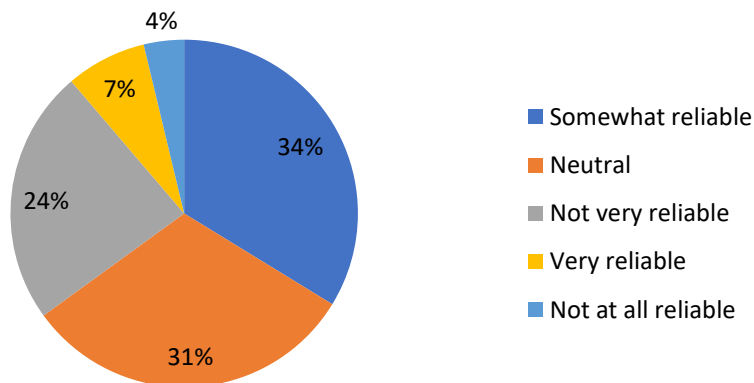


Figure 19 Opinion about the reliability of AI tools in detecting plagiarism or academic misconduct

In terms of readiness and training gaps: A significant number of respondents, 31%, saying they feel "slightly prepared" and 26% "not at all prepared", this indicates that even though AI is being used, many educators lack sufficient training or confidence in its broader application. So, with lack of Training consistently highlighted as the most significant challenge. The low preparedness ratings underline an urgent need for professional development programs in AI literacy. Institutions must prioritize ethical AI literacy programs to empower educators.

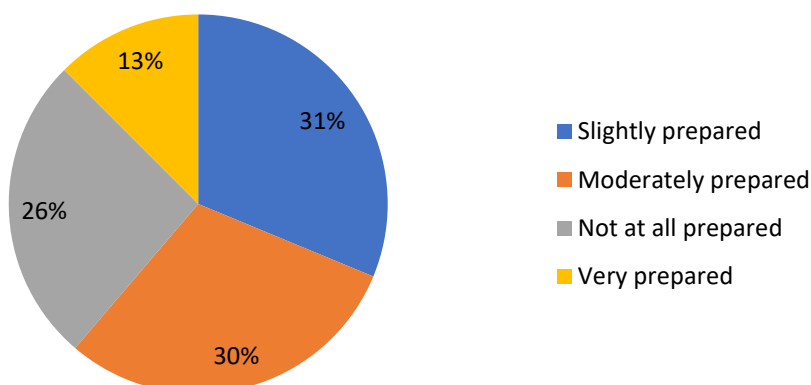


Figure 20 To what extent teachers feel prepared to guide students in the responsible and ethical use of AI tools

The most effective strategies to raise students' awareness of ethical AI risks (implied by high responses) are Interactive workshops and case studies are likely the top strategies, Lectures on ethics and guest speakers may follow, but with lower engagement potential (Figure 21). Students learn about ethical AI use in a better way through active participation (e.g., debates, real-world scenarios), than by passive methods (e.g., lectures), suggesting that curricula should prioritize hands-on learning. Institutions should integrate ethics into AI tool training rather than treating it as a standalone topic.

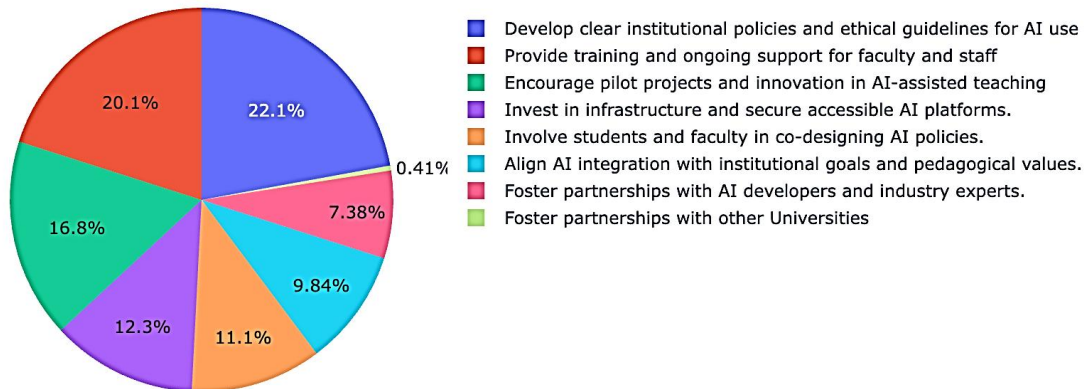


Figure 21 What Most effective strategies to raise students' awareness of the ethical risks associated with the use of AI

3.3.1 Strategies to Balance AI and Human Judgment in Education

Human-in-the-loop approaches (e.g., faculty reviewing AI outputs) are likely dominant, Clear guidelines on when to use AI vs. human input may also rank highly. Educators want AI to augment, not replace, their expertise—especially in subjective tasks (e.g., grading essays) and institutions must define boundaries (e.g., "AI drafts feedback, but professors finalize it") to alleviate fears of deskilling (Figure 22).

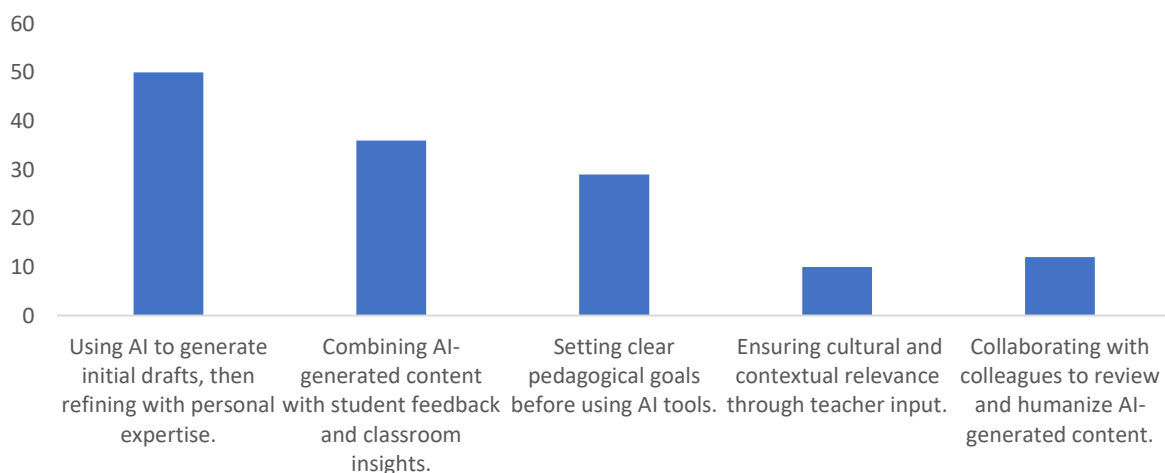


Figure 22 What strategies can higher education educators use to balance the use of AI and human judgement in developing educational resources

In terms of challenges in guiding students on ethical AI use the lack of training and unclear policies are likely the top barriers. Student over-reliance on AI (e.g., ChatGPT for essays) may also be a concern. On top of this, teachers don't feel confident to teach ethical AI use due to rapid tool evolution and absent institutional support (Figure 23). Solutions require both top-down (policy) and bottom-up (student workshops) interventions.

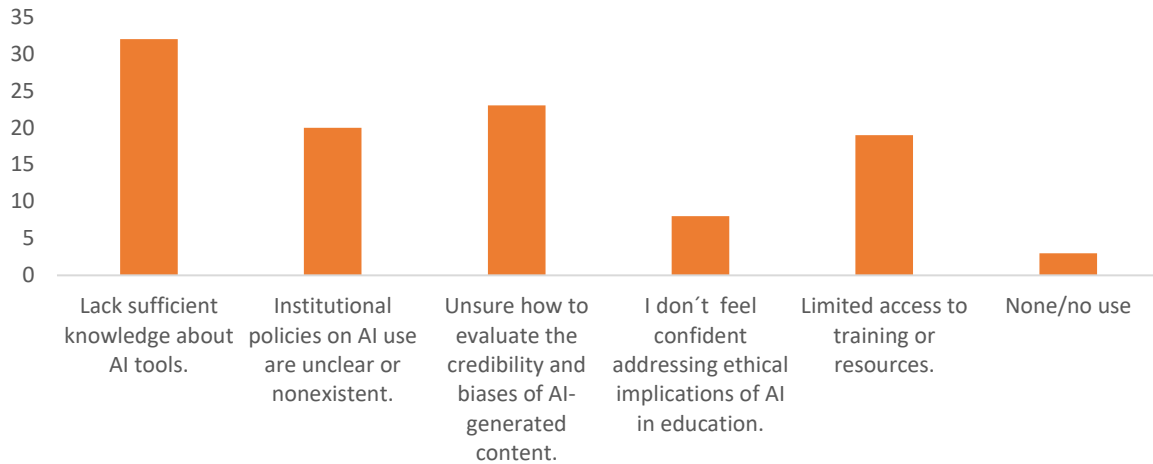


Figure 23 Challenges to guide students on the responsible and ethical use of AI

Main challenges when using AI Tools in teaching are the technical difficulties (e.g., tool complexity) and time constraints dominate. Data privacy and bias in AI outputs are also rank highly (Figure 24). This means that teachers and educators even enthusiastic face practical hurdles, highlighting the need for simplified, user-friendly AI tools, dedicated time for faculty to experiment and train and trust issues (privacy/bias) requiring transparency from AI vendors and institutions.

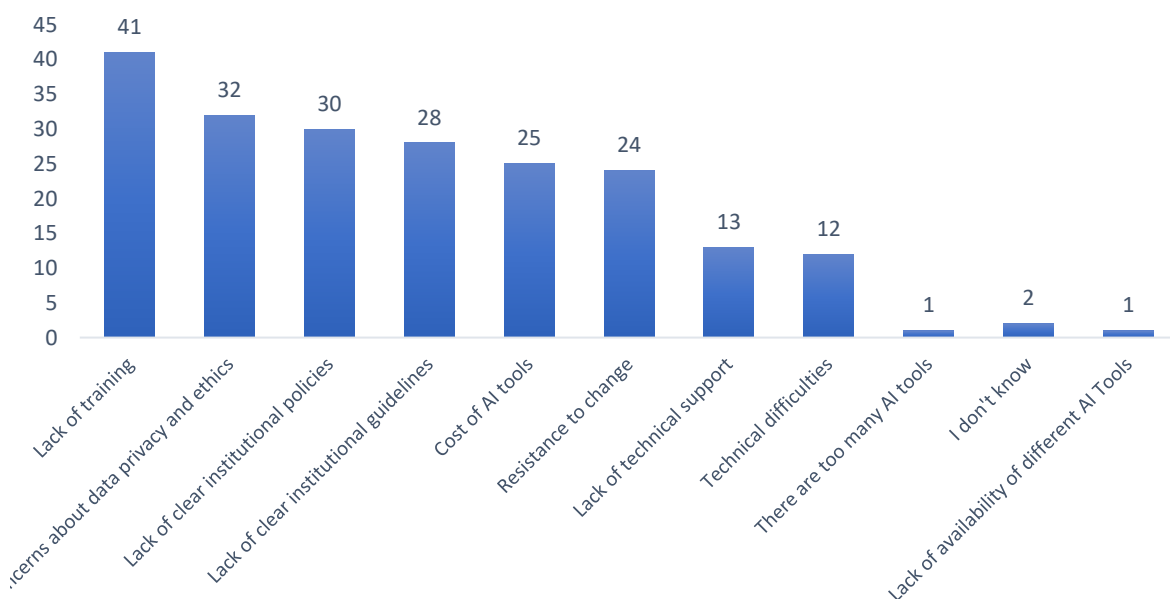


Figure 24- Main challenges higher education educators face when using AI tools in their teaching

Faculty scepticism primarily centres on ethical implications and data privacy. Before scale-up, institutions must not only invest in secure AI tools but also in transparency measures that keep both educators and students informed about how their data and content are used. The survey clearly shows that while there is recognition of AI's potential benefits for teaching and student support, it also comes with significant training and policy challenges. Proactively addressing these issues by developing comprehensive AI strategies, investing in training programs, and involving all stakeholders in decision-making can pave the way for more confident and effective AI adoption. The results indicate that the majority of the HE educator don't feel pressure of using AI tools (62%), this may be the reflect of the conservatism of the HE institutions and suggesting AI adoption is still optional or institutionally unenforced. This could slow integration unless accompanied by incentives or mandates (Figure 25).

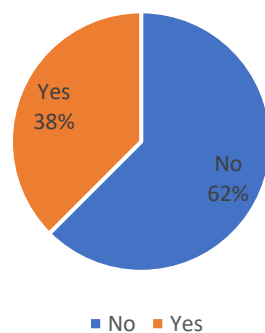


Figure 25- Answers to the question: Do you think higher education educators are under pressure to use AI tools in teaching?

This measured interest in participating in AI training implies that AI will become a transformative element in education only after institutions not only provide robust support and training but also address concerns around ethical use and transparency. By doing so, they can gradually shift from using AI primarily as an administrative aid to making it integral in real-time teaching, adaptive assessments, and personalized student interventions. Strong interest in training (high "yes" responses), as see in Figure 26, contrasts with low preparedness (Figure 20), revealing a readiness gap that institutions must fill to avoid ad-hoc, ineffective AI use.

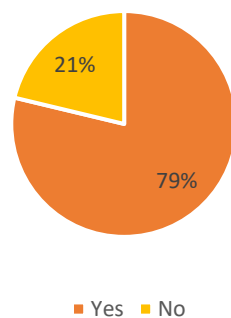


Figure 26 - Answers to the question: Would you be interested in participating in a specific training on the use of AI in teaching?

In terms of best formats for training the favourite types were webinars (49 responses) and in-person workshops (39 responses), signalling a desire for interactive learning environments. Online self-paced courses received fewer votes (7), suggesting that faculty favour real-time interaction alongside content delivery (Figure 27).

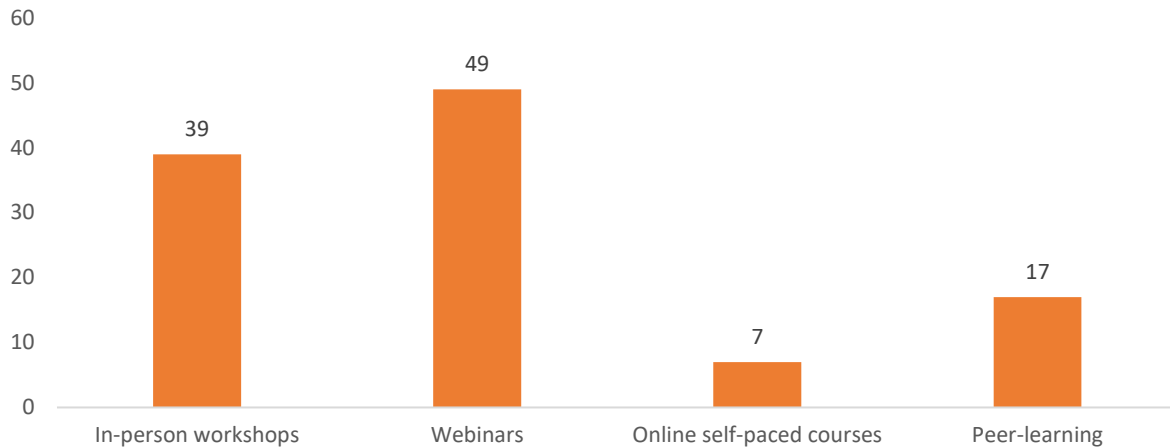


Figure 27 - Best formats of training to help higher education educators learn to use AI tools

Improved student outcomes, user-friendly tools, and clear guidelines are top drivers. Addressing these factors systematically (e.g., piloting intuitive tools) could accelerate adoption (Figure 28).

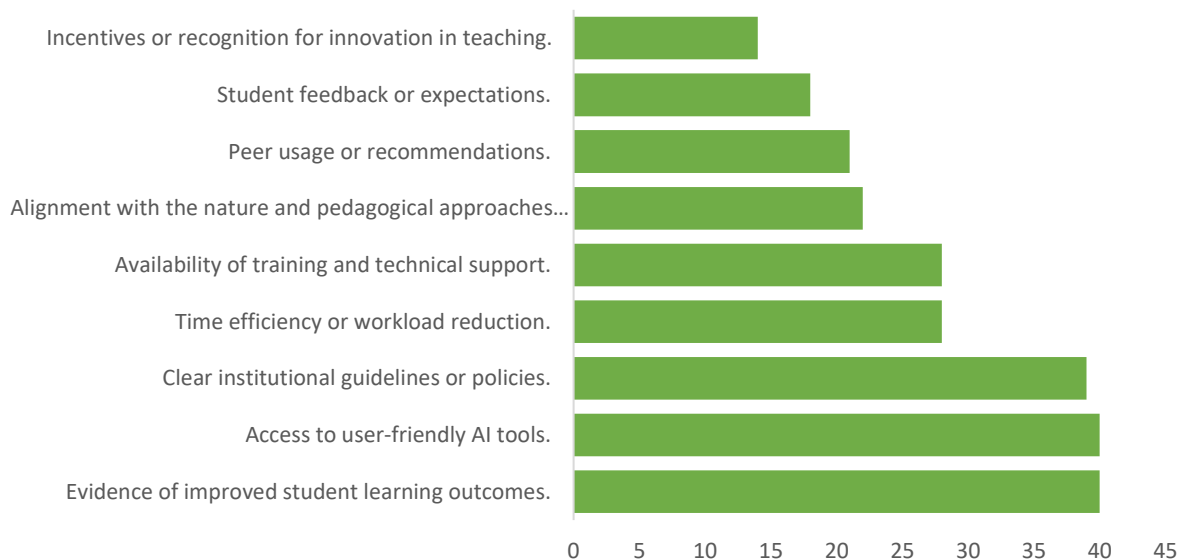


Figure 28 - Factors that most influence teacher decision to start or continue using AI in teaching

The best strategies to overcome resistance to AI adoption are through success stories (e.g., pilot programs showing improved outcomes) and incentives (e.g., funding/rewards). Peer mentoring may also be cited as effective (Figure 29).

Resistance comes from scepticism about AI's value. Tangible proof (e.g., "Professor X saved 10 hours/week using AI") can shift perceptions. Also, a culture of collaboration (e.g., AI "champions" in departments) reduces the fear of change.

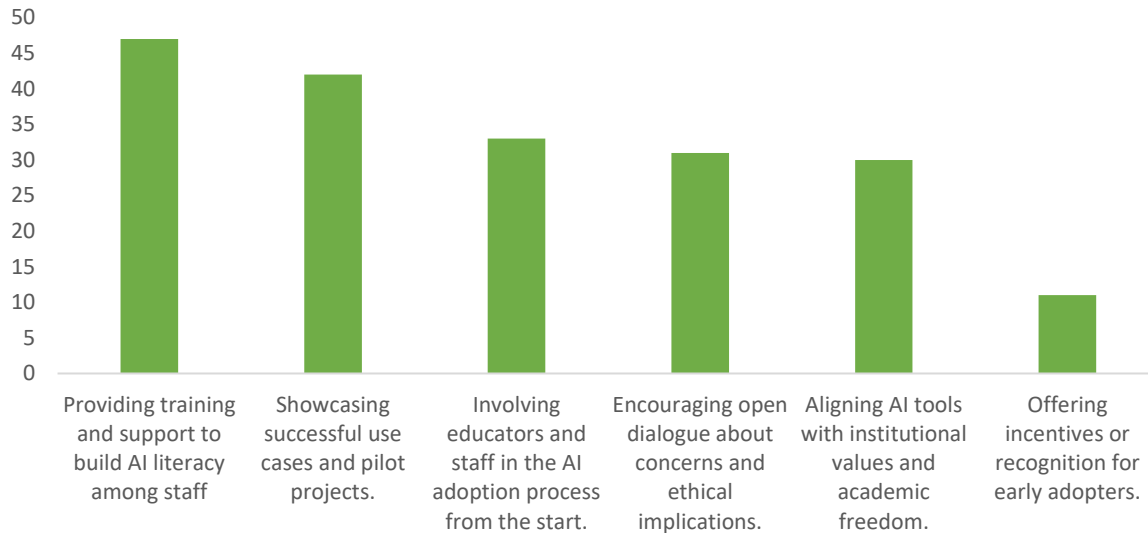


Figure 29 - Strategies that help to overcome cultural and change-resistance barriers to AI adoption in universities

The responses related with how AI Can Support Students with Special Needs, indicate that personalized learning (e.g., adaptive text-to-speech tools) and accessibility features (e.g., real-time captions) are preferred options. Reduced stigma (e.g., AI tutors for social anxiety) is also highlighted (Figure 30).

The AI's ability to customize support is a change factor for inclusivity, but institutions must train teachers to use these tools effectively and ensure AI solutions are vetted for accuracy (e.g., speech recognition for dyslexia).

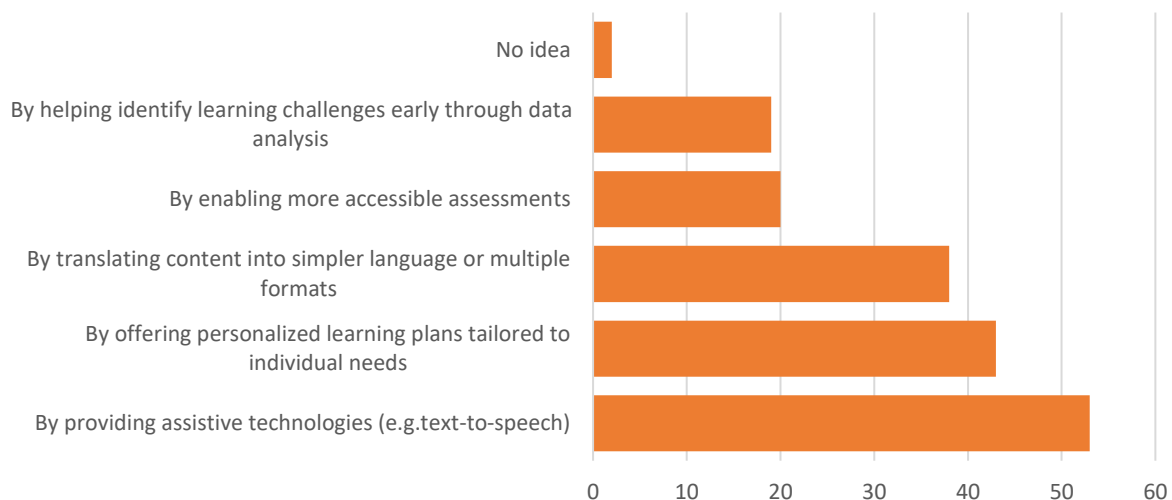


Figure 30 – Ways how AI can support students with special needs or learning difficulties

Faculty-led AI committees and optional tool adoption are probable top strategies for balancing AI implementation with academic autonomy. Transparent algorithms (e.g., explainable AI for grading) may also be valued (Figure 31).

Educators fear AI thinking it will undermine their authority. Involving them in decision-making (e.g., co-designing AI policies) builds trust. Optional vs. mandatory AI use should respect diverse teaching philosophies, and teachers' opinion, at the same time teachers should be encouraged to experiment it.

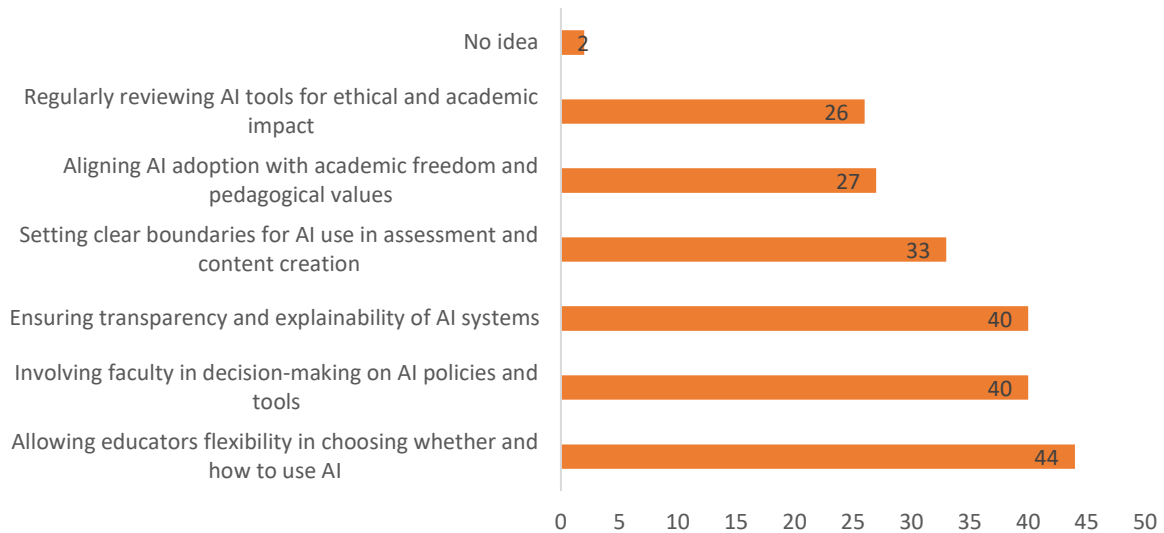


Figure 31 Ways how universities can balance the implementation of AI with preserving academic autonomy

Clear policies (54) and faculty training (49) are prioritized, reflecting a demand for structured support. Pilot projects (41) are also key, suggesting a preference for iterative, evidence-based scaling (Figure 32).

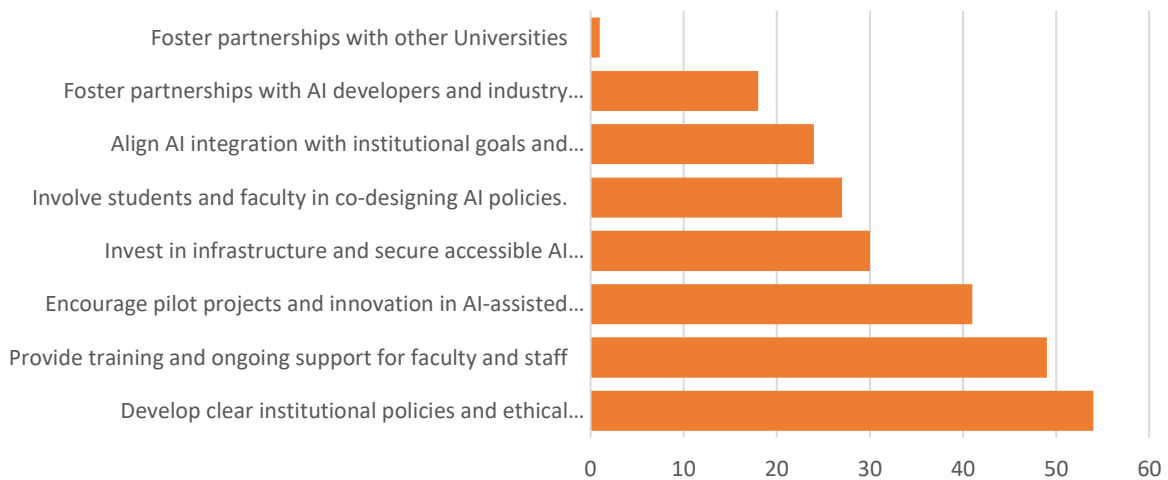


Figure 32 - Strategies that universities should adopt to effectively integrate AI into their educational system

3.3.2 Summary

The questionnaires results show that while there is clear enthusiasm for AI's potential in expanding and personalizing education, this optimism comes with a caveat—a strong need for human supervision and structured support. Also, these survey results reveal that while AI is gaining traction, there are systemic challenges hampering its full integration. The gaps in training and inconsistent policies mean that while experimental and preparatory use is underway, a uniform, institution-wide approach is necessary. For adoption to be effective, schools must establish clear guidelines, invest in training programs—including in-person workshops and webinars—and foster environments where faculty can learn by doing and share successful use cases.

3.4 Recommendations for a seamless transition

To fully harness AI's capabilities in educational processes, the survey suggests several actionable recommendations:

- **Develop Clear Institutional Policies:** Establish comprehensive ethical guidelines and operational standards to ensure that AI tools are implemented safely and effectively.
- **Invest in Robust Training Programs:** Implement targeted training—through webinars, workshops, and peer-learning sessions—that not only builds technical skills but also nurtures confidence in using AI across varying teaching stages.
- **Pilot Projects and Gradual Integration:** Start with pilot projects in course preparation and gradually extend AI usage into classroom interactions and assessments once the benefits are clearly demonstrated. This phased approach could help mitigate uncertainties and allow for adjustments based on direct feedback.
- **Enhance Communication and Student Involvement:** Improve institutional communication so that faculty receive regular updates and clear instructions on integrating AI. Involving students in this dialogue can also provide insights into how AI tools are enhancing their learning experience. Engaging students in AI ethics (e.g., via hackathons or feedback loops) could amplify impact.
- **Balance Automated and Human Oversight:** Ensuring that AI supplements rather than replaces human interaction will address reliability concerns and promote a balanced approach in teaching methodologies.
- **Regularly Review and Adapt AI Tools:** Create a mechanism to continually evaluate AI tool performance, technical support adequacy, and ethical implications to adapt quickly to new challenge

4. Conclusion

The survey outcomes reveal a dynamic yet cautious landscape for AI integration in higher education. Faculty recognize the transformative potential of AI for enhancing teaching and student engagement but also call for more robust training, clearer policies, and better technical and ethical safeguards. While tools like ChatGPT are popular in content generation and course preparation, trust in AI reliability remains moderate highlighting an opportunity for institutions to bridge the knowledge gap through targeted initiatives.

Moving forward, a balanced approach that combines human oversight with innovative AI applications will be essential to fully harness AI's capabilities while preserving academic values and ensuring student success.

In summary, while both AI and other educational technologies aim to support and enhance the learning process, AI distinguishes itself through its adaptive personalization, dynamic content generation, and real-time data analysis. However, these advanced capabilities come with higher complexity, demands for ethical scrutiny, and the need for robust infrastructure and human oversight. In contrast, traditional technologies offer reliable, well-understood benefits, primarily focused on content delivery and basic analytics, which make them easier to implement but less dynamic in optimizing the educational experience.

As educational institutions continue to evolve digitally, combining the strengths of AI with established technologies can create a balanced, innovative learning ecosystem that leverages deep personalization and efficiency while maintaining stability and clarity in teaching practices.